

18	(a)	Yes supported	P1	for process to find the surface area of the hemisphere, eg $4 \times \pi \times 3.2^2 \div 2$ (=64.33...)	
			P1	for process to find the curved surface area of the cone, eg $\pi \times 1.1 \times 8$ (27.64...)	
			P1	for process to consider area of circular face needing to be painted, eg $\pi \times 3.2^2 - \pi \times 1.1^2$ (=28.36...) OR for process to use coverage with one or more relevant area eg “64.33...” $\div 8.5$ (=7.56...) OR for process to work with coverage of 15 tins eg 8.5×15 (=127.5)	
			P1	for process to use coverage, eg (“64.33...” + “27.64...” + “28.36...”) $\div 8.5$ (=14.159...) OR (“64.33...” + “27.64...” + “28.36...” (=120.35..) and 8.5×15 (=127.5)	
	(b)	Explanation	C1	for conclusion “Yes” supported by accurate figures eg 14.1(59...) rounds to 15 OR 120(.35...) < 127.5	May be seen as each area $\div 8.5$ then combined 120.2934 using 3.14 for π
			C1	for valid explanation eg will increase the number of tins needed or Callum will not have enough tins of paint	14.152... using 3.14 for π